

A

 Omitted Proofs

592

593 ► **Lemma 9** (Splitting, improved formulation). *Let A , B , and K be formulas, and let a be an*
 594 *atom.*

1. *If there is a derivation* $\frac{\Phi \Vdash^{\text{BV}}}{(A \otimes B) \wp K}$, *then there are formulas K_A and K_B and derivations*

$$\frac{\frac{K_A \wp K_B}{\Phi_K \Vdash^{\text{BV}} K} \text{ and } \frac{\Phi_A \Vdash^{\text{BV}}}{A \wp K_A} \text{ and } \frac{\Phi_B \Vdash^{\text{BV}}}{B \wp K_B}, \text{ s.t. } \text{AF} \left(\frac{s \frac{\left(\frac{\Phi_A \Vdash}{A \wp K_A} \right) \otimes \left(\frac{\Phi_B \Vdash}{B \wp K_B} \right)}{(A \otimes B) \wp \left(\frac{K_A \wp K_B}{\Phi_K \Vdash K} \right)}}{(A \otimes B) \wp \left(\frac{K_A \wp K_B}{\Phi_K \Vdash K} \right)} \right) = \text{AF}(\Phi)$$

2. *If there is a derivation* $\frac{\Phi \Vdash^{\text{BV}}}{(A \triangleleft B) \wp K}$, *then there are formulas K_A and K_B and derivations*

$$\frac{\frac{K_A \triangleleft K_B}{\Phi_K \Vdash^{\text{BV}} K} \text{ and } \frac{\Phi_A \Vdash^{\text{BV}}}{A \wp K_A} \text{ and } \frac{\Phi_B \Vdash^{\text{BV}}}{B \wp K_B}, \text{ s.t. } \text{AF} \left(\frac{q \downarrow \frac{\left(\frac{\Phi_A \Vdash}{A \wp K_A} \right) \triangleleft \left(\frac{\Phi_B \Vdash}{B \wp K_B} \right)}{(A \triangleleft B) \wp \left(\frac{K_A \triangleleft K_B}{\Phi_K \Vdash K} \right)}}{(A \triangleleft B) \wp \left(\frac{K_A \triangleleft K_B}{\Phi_K \Vdash K} \right)} \right) = \text{AF}(\Phi)$$

3. *If there is a derivation* $\frac{\Phi \Vdash^{\text{BV}}}{a \wp K}$, *then there is a formula K and a derivation*

$$\frac{\frac{a^\perp}{\Phi_a \Vdash^{\text{BV}} K}, \text{ s.t. } \text{AF} \left(\frac{\text{ai} \downarrow \frac{\mathbb{I}}{a \wp \left(\frac{a^\perp}{\Phi_a \Vdash K} \right)}}{a \wp \left(\frac{a^\perp}{\Phi_a \Vdash K} \right)} \right) = \text{AF}(\Phi)$$

595 **Proof.** We recall that the standard proof of the splitting lemma proceeds by a case analysis
 596 on the last rule of the derivation Φ , and applying the induction hypothesis on the derivation
 597 with conclusion the premise of the last rule. The additional observation here is that each
 598 transformation preserves the atomic flow. This follows from the fact that the only rule in
 599 BV whose corresponding atomic flow is not an identity flow is the $\text{ai} \downarrow$ rule, and the only
 600 case in which it is involved is in the third item above, where the atomic flow is the same
 601 modulo the identity over flows given by the fact that string diagrams are considered up-to
 602 the equivalences in the right of Figure 3, in particular, the ones allowing us to freely ‘slide’
 603 boxes over and under wires.

604 1. If there is a derivation $\frac{\Phi \Vdash^{\text{BV}}}{(A \otimes B) \wp K}$, then we have three cases:

- the bottom-most rule is applied to K , that is $\Phi = \frac{\Phi' \Vdash}{(A \otimes B) \wp \left(r \frac{K'}{K} \right)}$, in which case

we can conclude by applying the induction hypothesis to Φ' , and then applying the

same rule instance r to the resulting derivation, obtaining derivations

$$\Phi_K = \left(\begin{array}{c} K_A \wp K_B \\ \Phi_{K'} \parallel \\ K' \\ r \\ K \end{array} \right) \quad \text{and} \quad \begin{array}{c} \Phi_A \parallel \\ A \wp K_A \end{array} \quad \text{and} \quad \begin{array}{c} \Phi_B \parallel \\ B \wp K_B \end{array}$$

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We conclude since the atomic flow of s is an identity flow, therefore we have

$$\text{AF} \left(\begin{array}{c} s \\ \left(\begin{array}{c} \Phi_A \parallel \\ A \wp K_A \end{array} \right) \otimes \left(\begin{array}{c} \Phi_B \parallel \\ B \wp K_B \end{array} \right) \\ \hline \left(\begin{array}{c} K_A \wp K_B \\ \Phi_{K'} \parallel \\ K' \\ r \\ K \end{array} \right) \\ (A \otimes B) \wp \end{array} \right) = \begin{array}{c} \text{AF}(\Phi_A) \quad \text{AF}(\Phi_B) \\ \text{AF}(\Phi_K) \\ \text{AF}(r) \end{array} = \begin{array}{c} \text{AF}(\Phi') \\ \text{AF}(r) \end{array} = \text{AF}(\Phi) \quad (17)$$

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- the bottom-most rule is a $\equiv$ applied to $A \otimes B$. In this case, we have that $A = C_{i_1} \otimes \dots \otimes C_{i_k}$ and $B = C_{j_1} \otimes \dots \otimes C_{j_m}$ for some $i_1, \dots, i_k, j_1, \dots, j_m$ such that $\{i_1, \dots, i_k, j_1, \dots, j_m\} = \{1, \dots, n\}$, and that

$$\Phi = \left(\begin{array}{c} \Phi' \parallel \\ \left(\begin{array}{c} C'_1 \\ \dots \\ C_1 \end{array} \right) \otimes \dots \otimes \left(\begin{array}{c} C'_n \\ \dots \\ C_n \end{array} \right) \\ \hline \left(\begin{array}{c} C_{i_1} \otimes \dots \otimes C_{i_k} \\ = \\ A \end{array} \right) \otimes \left(\begin{array}{c} C_{j_1} \otimes \dots \otimes C_{j_m} \\ = \\ B \end{array} \right) \\ \hline \end{array} \right) \wp K$$

for some $C'_k \equiv C_k$ for all $k \in \{1, \dots, n\}$. In this case, we repetively apply splitting to the conclusion of Φ' , until we get derivations

$$\begin{array}{c} K_1 \wp \dots \wp K_n \\ \Phi_K \parallel \\ K \end{array} \quad \text{and} \quad \begin{array}{c} \Phi_{C_k} \parallel \\ C'_k \wp K_k \end{array} \quad \text{for all } k \in \{1, \dots, n\}$$

And conclude since we can reconstruct

$$\Phi_A = s \frac{\left(\begin{array}{c} \Phi_{C_{i_1}} \parallel \\ C'_{i_1} \wp K_{i_1} \end{array} \right) \otimes \dots \otimes \left(\begin{array}{c} \Phi_{C_{i_k}} \parallel \\ C'_{i_k} \wp K_{i_k} \end{array} \right)}{\left(\begin{array}{c} C'_{i_1} \otimes \dots \otimes C'_{i_k} \\ \equiv \\ A \end{array} \right) \wp K_{i_1} \wp \dots \wp K_{i_k}}$$

and

$$\Phi_B = s \frac{\left(\begin{array}{c} \Phi_{C_{j_1}} \parallel \\ C'_{j_1} \wp K_{j_1} \end{array} \right) \otimes \dots \otimes \left(\begin{array}{c} \Phi_{C_{j_m}} \parallel \\ C'_{j_m} \wp K_{j_m} \end{array} \right)}{\left(\begin{array}{c} C'_{j_1} \otimes \dots \otimes C'_{j_m} \\ \equiv \\ B \end{array} \right) \wp K_{j_1} \wp \dots \wp K_{j_m}}$$

because we have that

$$\text{AF} \left(s \frac{\left(\frac{\Phi_A \parallel}{A \wp K_A} \right) \otimes \left(\frac{\Phi_B \parallel}{B \wp K_B} \right)}{(A \otimes B) \wp \left(\frac{K_A \wp K_B}{\Phi_K \parallel K} \right)} \right) = \begin{array}{c} \text{AF}(\Phi_{C_1} \otimes \dots \otimes \Phi_{C_n}) \\ \text{---} \\ \begin{array}{c} C'_1 \quad K_{C_1} \quad \dots \quad C'_n \quad K_{C_n} \\ \text{---} \\ \text{AF}(\equiv) \quad \text{AF}(\Phi_K) \\ \text{---} \\ C_1 \quad \dots \quad C_n \quad K \end{array} \end{array} = \text{AF}(\Phi)$$

607 where $K_A = K_{i_1} \wp \dots \wp K_{i_k}$ and $K_B = K_{j_1} \wp \dots \wp K_{j_m}$;

- or the bottom-most rule in Φ is a s , that is $\Phi = \frac{\Phi' \parallel}{s \frac{A \otimes (B \wp K)}{(A \otimes B) \wp K}}$, in which case we can apply inductive hypothesis to Φ' , obtaining derivations

$$\frac{K_B}{\Phi_K \parallel K} \quad \text{and} \quad \frac{\Phi_A \parallel}{A} \quad \text{and} \quad \frac{\Phi_B \parallel}{B \wp K_B}$$

and conclude immediately since

$$\text{AF} \left(s \frac{\left(\frac{\Phi_A \parallel}{A} \right) \otimes \left(\frac{\Phi_B \parallel}{B \wp K_B} \right)}{(A \otimes B) \wp \left(\frac{K_B}{\Phi_K \parallel K} \right)} \right) = \text{AF} \left(s \frac{\left(\frac{\Phi_A \parallel}{A} \right) \otimes \left(B \wp \left(\frac{K_B}{\Phi_K \parallel K} \right) \right)}{(A \otimes B) \wp K} \right) = \text{AF}(\Phi)$$

608 because the atomic flow of s is an identity flow.

- 609 2. If there is a derivation $\frac{\Phi \parallel \text{BV}}{(A \triangleleft B) \wp K}$, then we conclude similarly to the previous case.

610 The only difference is that in this case we have to consider the rule $\mathbf{q}\downarrow$ instead of the
611 rule s , whose atomic flow is not an identity flow, but it is constructed by identities and
612 crossings.

- 613 3. If there is a derivation $\frac{\Phi \parallel \text{BV}}{a \wp K}$, then we have two cases:

- either the last rule is applied to K , that is $\Phi = \frac{\Phi' \parallel}{a \wp \left(r \frac{K'}{K} \right)}$, in which case we can

conclude by applying the induction hypothesis to Φ' , and then applying the same

rule instance r to the resulting derivation, obtaining a derivation $\Phi_a = \frac{a^\perp}{\Phi'_a \parallel \frac{K'}{r \frac{K}{K}}}$. We

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conclude since

$$\text{AF} \left(\text{ai} \downarrow \frac{\mathbb{I}}{a \wp \left(\begin{array}{c} a^\perp \\ \Phi'_a \parallel \\ r \frac{K'}{K} \end{array} \right)} \right) = \text{AF} \left(\text{ai} \downarrow \frac{\mathbb{I}}{a \wp \left(\begin{array}{c} a^\perp \\ \Phi'_a \parallel \\ K' \end{array} \right)} \right) = \text{AF} \left(\begin{array}{c} \text{AF}(\Phi') \\ \dots \\ \text{AF}(r) \\ \dots \end{array} \right) = \text{AF}(\Phi)$$

■ or the last rule is applied to a , in which case it must be a $\text{ai} \downarrow$ and $\Phi = \frac{\Phi' \parallel K'}{\left(r \frac{\mathbb{I}}{a \wp a^\perp} \right) \wp K'}$

In this case, we conclude by letting $\Phi_a = \frac{a^\perp \wp \left(\begin{array}{c} \Phi' \parallel \\ K' \end{array} \right)}{K}$ since

$$\text{AF} \left(\text{ai} \downarrow \frac{\mathbb{I}}{a \wp \left(\begin{array}{c} a^\perp \\ \Phi_a \parallel \\ K \end{array} \right)} \right) = \text{af} \left(\begin{array}{c} \text{AF}(\Phi') \\ \dots \end{array} \right) = \text{AF}(\Phi)$$

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615 ► **Lemma 11.** *If there is a derivation $\Phi = C \left[\frac{\Phi' \parallel \text{BV}}{\text{q} \uparrow \frac{(A \triangleleft B) \otimes (C \triangleleft D)}{(A \otimes C) \triangleleft (B \otimes D)}} \right]$ then there is a*

616 *derivation $\frac{\Phi^* \parallel \text{BV}}{C[(A \otimes C) \triangleleft (B \otimes D)]}$ such that $\text{AF}(\Phi^*) = \text{AF}(\Phi)$.*

Proof. We start by applying context reduction to Φ' , obtaining a formula K and derivations

$$\frac{\Phi'_1 \parallel \text{BV}}{((A \otimes C) \triangleleft (B \otimes D)) \wp K} \quad \text{and} \quad \frac{X \wp K}{\Phi'_{C[X]} \parallel \text{BV}} \quad \text{for every formula } X$$

617 such that

$$\text{AF}(\Phi') = \frac{\text{AF}(\Phi'_1)}{\text{AF}(\Phi'_{C[(A \otimes C) \triangleleft (B \otimes D)]})} \quad (18)$$

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We then apply the splitting lemma to Φ' , obtaining formulas K_l and K_r and derivations

$$\frac{K_l \wp K_r}{\Phi_K \parallel \text{BV}} \quad \text{and} \quad \frac{\Phi_{A \triangleleft B} \parallel \text{BV}}{(A \triangleleft B) \wp K_l} \quad \text{and} \quad \frac{\Phi_{C \triangleleft D} \parallel \text{BV}}{(C \triangleleft D) \wp K_r}$$

such that

$$\text{AF}(\Phi') = \text{AF}\left(s \frac{\Phi_{A \triangleleft B} \otimes \Phi_{C \triangleleft D}}{((A \triangleleft B) \otimes (C \triangleleft D)) \wp \Phi_K}\right) = \begin{array}{c} \text{AF}(\Phi_{A \triangleleft B}) \quad \text{AF}(\Phi_{C \triangleleft D}) \\ \text{---} \quad \text{---} \\ \text{AF}(\Phi_K) \\ \dots \end{array} \quad (19)$$

Then, we can apply splitting on both $\Phi_{A \triangleleft B}$ and $\Phi_{C \triangleleft D}$, obtaining formulas K_A, K_B, K_C , and K_D and derivations

$$\begin{array}{c} K_A \triangleleft K_B \\ \Phi_l \parallel \text{BV} \\ K_l \end{array}, \quad \begin{array}{c} K_C \triangleleft K_D \\ \Phi_r \parallel \text{BV} \\ K_r \end{array}, \quad \begin{array}{c} \Phi_A \parallel \text{BV} \\ A \wp K_A \end{array}, \quad \begin{array}{c} \Phi_B \parallel \text{BV} \\ B \wp K_B \end{array}, \quad \begin{array}{c} \Phi_C \parallel \text{BV} \\ C \wp K_C \end{array} \quad \text{and} \quad \begin{array}{c} \Phi_D \parallel \text{BV} \\ D \wp K_D \end{array}$$

such that

$$\text{AF}(\Phi_{A \triangleleft B}) = \text{AF}(\Phi_{A \triangleleft B}) = \text{AF}\left(q\downarrow \frac{\Phi_A \triangleleft \Phi_B}{(A \triangleleft B) \wp \Phi_l}\right) = \begin{array}{c} \text{AF}(\Phi_A) \quad \text{AF}(\Phi_B) \\ \text{---} \quad \text{---} \\ \text{AF}(\Phi_l) \\ \dots \end{array} \quad (20)$$

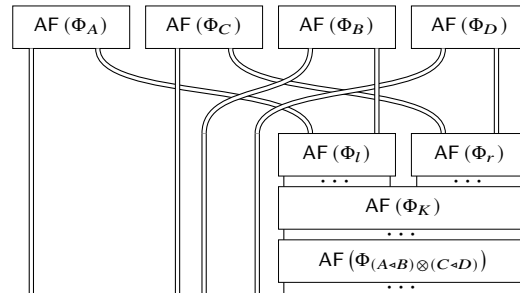
and

$$\text{AF}(\Phi_{C \triangleleft D}) = \text{AF}(\Phi_{C \triangleleft D}) = \text{AF}\left(q\downarrow \frac{\Phi_C \triangleleft \Phi_D}{(C \triangleleft D) \wp \Phi_r}\right) = \begin{array}{c} \text{AF}(\Phi_C) \quad \text{AF}(\Phi_D) \\ \text{---} \quad \text{---} \\ \text{AF}(\Phi_r) \\ \dots \end{array} \quad (21)$$

We can now build the derivation Φ^* as shown below.

$$\begin{array}{c} \left(s \frac{\Phi_A \otimes \Phi_C}{(A \otimes C) \wp K_A \wp K_C} \right) \triangleleft \left(s \frac{\Phi_B \otimes \Phi_D}{(B \otimes D) \wp K_B \wp K_D} \right) \\ q\downarrow \frac{\quad}{((A \otimes C) \triangleleft (B \otimes D)) \wp \left(\begin{array}{c} \Phi_l \wp \Phi_r \\ \dots \\ \Phi_K \end{array} \right)} \\ \Phi'_{C[(A \otimes C) \triangleleft (B \otimes D)]} \parallel \\ C[(A \otimes C) \triangleleft (B \otimes D)] \end{array}$$

Whose atomic flow below is such that $\text{AF}(\Phi^*) = \text{AF}(\Phi)$ by Equations (19)–(21) and Figure 3.



B

 Definition of BV-category

In this appendix we give the definition of a BV-category with negation in the sense of [6].

► **Definition 30** (Normal Duoidal Monoidal Structures). *Let $\mathbf{C}$ be a category with two monoidal structures $(\mathbf{C}, I, \otimes)$ and $(\mathbf{C}, J, \lhd)$. We say that $(J, \lhd)$ is normal duoidal to $(I, \otimes)$ if we have a natural transformation:*

$$w : (A \lhd B) \otimes (C \lhd D) \rightarrow (A \otimes C) \lhd (B \otimes D)$$

called weak interchange, together with morphisms

$$i : I \cong J \quad \delta_I : I \rightarrow I \lhd I \quad \mu_J : J \otimes J \rightarrow J$$

which satisfy the following coherence properties:

the weak interchange respects associativity in the sense that the following diagrams

$$\begin{array}{ccc}
 ((A \lhd B) \otimes (C \lhd D)) \otimes (E \lhd F) & \xrightarrow{\alpha} & (A \lhd B) \otimes ((C \lhd D) \otimes (E \lhd F)) \\
 \downarrow w \otimes \text{id} & & \downarrow \text{id} \otimes w \\
 ((A \otimes C) \lhd (B \otimes D)) \otimes (E \lhd F) & & (A \lhd B) \otimes ((C \otimes E) \lhd (D \otimes F)) \\
 \downarrow w & & \downarrow w \\
 ((A \otimes C) \otimes E) \lhd ((B \otimes D) \otimes F) & \xrightarrow{\alpha \lhd \alpha} & (A \otimes (C \otimes E)) \lhd (B \otimes (D \otimes F))
 \end{array} \tag{22}$$

$$\begin{array}{ccc}
 ((A \lhd C) \lhd E) \otimes ((B \lhd D) \lhd F) & \xrightarrow{\alpha \otimes \alpha} & (A \lhd (C \lhd E)) \otimes (B \lhd (D \lhd F)) \\
 \downarrow w & & \downarrow w \\
 ((A \lhd C) \otimes (B \lhd D)) \lhd (E \otimes F) & & (A \otimes B) \lhd ((C \lhd E) \otimes (D \lhd F)) \\
 \downarrow w \lhd \text{id} & & \downarrow \text{id} \lhd w \\
 ((A \otimes B) \lhd (C \otimes D)) \lhd (E \otimes F) & \xrightarrow{\alpha} & (A \otimes B) \lhd ((C \otimes D) \lhd (E \otimes F))
 \end{array} \tag{23}$$

commute; the weak interchange respects unitality in the sense that the following diagrams

$$\begin{array}{ccc}
 I \otimes (A \lhd B) & \xrightarrow{\delta_I \otimes \text{id}} & (I \lhd I) \otimes (A \lhd B) \\
 \lambda_{A \lhd B} \uparrow & & \downarrow w \\
 A \lhd B & \xrightarrow{\lambda_A \lhd \lambda_B} & (I \otimes A) \lhd (I \otimes B)
 \end{array}
 \quad
 \begin{array}{ccc}
 (A \lhd B) \otimes I & \xrightarrow{\text{id} \otimes \delta_I} & (A \lhd B) \otimes (I \lhd I) \\
 \rho_{A \lhd B} \uparrow & & \downarrow w \\
 A \lhd B & \xrightarrow{\rho_A \lhd \rho_B} & (A \otimes I) \lhd (B \otimes I)
 \end{array} \tag{24}$$

$$\begin{array}{ccc}
 J \lhd (A \otimes B) & \xleftarrow{\mu_J \lhd \text{id}} & (J \otimes J) \lhd (A \otimes B) \\
 \lambda_{A \otimes B} \uparrow & & \uparrow w \\
 A \otimes B & \xrightarrow{\lambda_A \otimes \lambda_B} & (J \lhd A) \otimes (J \lhd B)
 \end{array}
 \quad
 \begin{array}{ccc}
 (A \otimes B) \lhd J & \xleftarrow{\text{id} \lhd \mu_J} & (A \otimes B) \lhd (J \otimes J) \\
 \rho_{A \otimes B} \uparrow & & \uparrow w \\
 A \otimes B & \xrightarrow{\rho_A \otimes \rho_B} & (A \lhd J) \otimes (B \lhd J)
 \end{array} \tag{25}$$

commute; the interchange commutes with the symmetry of $\otimes$ in the sense that the following diagram

$$\begin{array}{ccc}
 (A \triangleleft B) \otimes (C \triangleleft D) & \xrightarrow{\cong} & (C \triangleleft D) \otimes (A \triangleleft B) \\
 \downarrow w & & \downarrow w \\
 (A \otimes C) \triangleleft (B \otimes D) & \xrightarrow{\cong} & (C \otimes A) \triangleleft (D \otimes B)
 \end{array} \tag{26}$$

commutes; we furthermore require that the isomorphism i is an isomix map, i.e. the following diagram

$$\begin{array}{ccc}
 J \otimes J & \xrightarrow{\text{id} \otimes i^{-1}} & J \otimes I \\
 \downarrow i^{-1} \otimes \text{id} & & \downarrow \cong \\
 I \otimes J & \xrightarrow{\cong} & J
 \end{array}$$

commutes; we also require that (I, i, δ_I) is a comonoid object with respect to the $(C, J, \triangleleft)$ monoidal structure and we require that (J, i^{-1}, μ_J) is a monoid object with respect to the $(C, I, \otimes)$ monoidal structure.

► **Definition 31.** A BV-category with negation is a $\ast$ -autonomous category $(C, I, \otimes, \multimap)$ with an additional monoidal structure $(C, J, \triangleleft)$, such that

■ $(C, I, \otimes)$ is normal duoidal to $(C, J, \triangleleft)$ and we write

$$w: (A \triangleleft B) \otimes (C \triangleleft D) \rightarrow (A \otimes C) \triangleleft (B \otimes D)$$

for the weak interchange natural transformation;

■ $(C, J, \triangleleft)$ is normal duoidal to $(C, \perp, \wp)$ and we write

$$w': (A \wp B) \triangleleft (C \wp D) \rightarrow (A \triangleleft C) \wp (B \triangleleft D)$$

for the weak interchange natural transformation;

and such that the following diagram

$$\begin{array}{ccc}
 (A \triangleleft B) \otimes ((C \wp E) \triangleleft (D \wp F)) & \xrightarrow{w} & (A \otimes (C \wp E)) \triangleleft (B \otimes (D \wp F)) \\
 \downarrow \text{id} \otimes w' & & \downarrow a \triangleleft a \\
 (A \triangleleft B) \otimes ((C \triangleleft D) \wp (E \triangleleft F)) & & ((A \otimes C) \wp E) \triangleleft ((B \otimes D) \wp F) \\
 \downarrow a & & \downarrow w' \\
 ((A \triangleleft B) \otimes (C \triangleleft D)) \wp (E \triangleleft F) & \xrightarrow{w \wp \text{id}} & ((A \otimes C) \triangleleft (B \otimes D)) \wp (E \triangleleft F)
 \end{array} \tag{27}$$

and its symmetric dual commute.

This definition implies that the $\ast$ -autonomous structure of C is isomix.

C Proof of Theorem 18

► **Theorem 18.** Let $\langle C, \otimes, \triangleleft, \mathbb{I}, \cdot^\perp, U \rangle$ be a strong BV-category. Then $\langle C, \otimes, \triangleleft, \mathbb{I}, \cdot^\perp, U \rangle$ is also a BV-category (with negation) in the sense of [6] and [25].

Proof. The proof follows easily from the fact that U is faithful and strictly preserves all the relevant data which allows us to reduce the commutativity of all the relevant diagrams in

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$\mathbf{C}$ to their commutativity in $\mathbf{D}$ under U . The latter is very easy to verify using the string diagram language (or coherence theorem) for strict compact closed categories.

To illustrate this, let us show why diagram (22) commutes in $\mathbf{C}$. Since U is faithful and strictly preserves the two monoidal structures in (22), it follows that diagram (22) commutes in $\mathbf{C}$, i.e.

$$\begin{array}{ccc}
 ((A \triangleleft B) \otimes (C \triangleleft D)) \otimes (E \triangleleft F) & \xrightarrow{\alpha} & (A \triangleleft B) \otimes ((C \triangleleft D) \otimes (E \triangleleft F)) \\
 q' \otimes \text{id} \downarrow & & \downarrow \text{id} \otimes q' \\
 ((A \otimes C) \triangleleft (B \otimes D)) \otimes (E \triangleleft F) & & (A \triangleleft B) \otimes ((C \otimes E) \triangleleft (D \otimes F)) \\
 q' \downarrow & & \downarrow q' \\
 ((A \otimes C) \otimes E) \triangleleft ((B \otimes D) \otimes F) & \xrightarrow{\alpha \triangleleft \alpha} & (A \otimes (C \otimes E)) \triangleleft (B \otimes (D \otimes F))
 \end{array}$$

commutes in $\mathbf{C}$, iff the diagram

$$\begin{array}{ccc}
 UA \boxtimes UB \boxtimes UC \boxtimes UD \boxtimes UE \boxtimes UF & \xrightarrow{\text{id}} & UA \boxtimes UB \boxtimes UC \boxtimes UD \boxtimes UE \boxtimes UF \\
 \text{id} \boxtimes \sigma \boxtimes \text{id} \boxtimes \text{id} \downarrow & & \downarrow \text{id} \boxtimes \text{id} \boxtimes \sigma \boxtimes \text{id} \\
 UA \boxtimes UC \boxtimes UB \boxtimes UD \boxtimes UE \boxtimes UF & & UA \boxtimes UB \boxtimes UC \boxtimes UE \boxtimes UD \boxtimes UF \\
 \text{id} \boxtimes \sigma \boxtimes \text{id} \downarrow & & \downarrow \text{id} \boxtimes \sigma \boxtimes \text{id} \\
 UA \boxtimes UC \boxtimes UE \boxtimes UB \boxtimes UD \boxtimes UF & \xrightarrow{\text{id}} & UA \boxtimes UC \boxtimes UE \boxtimes UB \boxtimes UD \boxtimes UF
 \end{array}$$

commutes in $\mathbf{D}$ (which we recall is *strict* compact closed) and this follows immediately from the coherence theorem of compact closed categories (in fact also from the coherence theorem for symmetric monoidal categories). In the upper diagram, the natural transformation q' is defined in the same way as in Definition 19 for the $q\uparrow$ rule.

The remaining diagrams can be checked in a similar way using the same argument, i.e. the fact that U is faithful and strictly preserves all the relevant data and then by the coherence theorem for compact closed categories. $\blacktriangleleft$

D Proof of Lemma 21

Lemma 32. For a SBV-derivation $\Phi = \frac{A}{\Phi \parallel \text{SBV}}, \frac{B}{\phantom{\Phi \parallel \text{SBV}}}$, the following diagram commutes (in $\mathbf{D}$).

$$\begin{array}{ccc}
 U\llbracket A \rrbracket & \equiv & \langle A \rangle \\
 U\llbracket \Phi \rrbracket \downarrow & & \downarrow \langle \Phi \rangle \\
 U\llbracket B \rrbracket & \equiv & \langle B \rangle
 \end{array} \tag{28}$$

Proof. This follows by induction on Φ . The two most interesting cases are $q\downarrow$ and $q\uparrow$. The case for $q\downarrow$ follows immediately from the definition of a strong BV-category. The case for $q\uparrow$ follows easily, because $U\llbracket q\uparrow \rrbracket = U(\cong) \circ U(\llbracket q \rrbracket^\perp) \circ U(\cong) = \text{id} \circ (\text{id} \boxtimes \sigma^\perp \boxtimes \text{id}) \circ \text{id} = \text{id} \boxtimes \sigma \boxtimes \text{id} = \langle q\uparrow \rangle$ where we used the fact that U strictly preserves all the relevant structure and maps the two unnamed isomorphisms to identities in $\mathbf{D}$.

The remaining cases are straightforward and follow easily from standard results about *-autonomous and compact closed categories. We illustrate this by considering the $ai\uparrow$ rule.

In order to prove this, recall that a compact closed category $\mathbf{D}$ is symmetric monoidal closed with internal hom $A \multimap B = B \otimes A^\perp$ and evaluation morphism $\text{ev}'_{A,B}: (A \multimap B) \otimes A \rightarrow B$ defined by

$$\text{ev}'_{A,B} = \left((A \multimap B) \otimes A = B \otimes A^\perp \otimes A \xrightarrow{\text{id} \otimes \sigma} B \otimes A \otimes A^\perp \xrightarrow{\text{id} \otimes \epsilon_A} B \right),$$

where $\epsilon_A: A \otimes A^\perp \rightarrow \mathbb{I}$ is the counit of the compact closed structure. By definition, we have that

$$\left[\text{ai} \uparrow \frac{a \otimes a^\perp}{\mathbb{I}} \right] = \llbracket a \rrbracket \otimes \llbracket a \rrbracket^\perp \xrightarrow{\sigma} \llbracket a \rrbracket^\perp \otimes \llbracket a \rrbracket \xrightarrow{\text{ev}} \mathbb{I}.$$

Applying U to this definition, we get

$$\begin{aligned} U(\text{ev}) \circ U(\sigma) &= \text{ev}' \circ \sigma && (U \text{ strictly preserves } *- \text{autonomous structure}) \\ &= (\text{id}_{\mathbb{I}} \otimes \epsilon) \circ (\text{id}_{\mathbb{I}} \otimes \sigma) && (\text{Definition}) \\ &= \epsilon \circ \sigma \circ \sigma && (\mathbf{D} \text{ is strict monoidal}) \\ &= \epsilon && (\text{Property of symmetry}) \\ &= \left(\text{ai} \uparrow \frac{a \otimes a^\perp}{\mathbb{I}} \right) && (\text{Definition}) \end{aligned}$$

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E Proof of Theorem 27

► **Theorem 27.** *The tuple $\langle \text{Caus}[\mathbf{CP}^*[\mathbf{FHilb}_{\text{sk}}]], \otimes, <, \mathbb{I}, \cdot^*, U \rangle$ is a strong BV-category.*

Proof. The category $\mathbf{FHilb}_{\text{sk}}$ is compact closed, because it is categorically equivalent to $\mathbf{Mat}_{\mathbb{C}}$ from [26] which is shown to be compact closed in [26]. Moreover, it is easy to see that it is *strict* compact closed by examining the relevant data in [26]. Furthermore, it is straightforward to see that it is *dagger* compact closed (with dagger given by the conjugate transpose of matrices) because it is the skeletal subcategory of $\mathbf{FHilb}$ which is well-known to be dagger compact closed [26]. The $\mathbf{CP}^*[-]$ construction (see [26, Chapter 7]) preserves dagger compact closure [26, Proposition 7.26], so it follows that $\mathbf{CP}^*[\mathbf{FHilb}_{\text{sk}}]$ is also dagger compact closed. Furthermore, $\mathbf{CP}^*[\mathbf{FHilb}_{\text{sk}}]$ is a strict symmetric monoidal category [26, Proposition 7.22]. The double dual isomorphism in $\mathbf{CP}^*[\mathbf{FHilb}_{\text{sk}}]$ is the identity, which can be easily proven using [26, Proposition 7.24]. Moreover, we also have that $(A \boxtimes B)^\perp = A^\perp \boxtimes B^\perp$ in $\mathbf{CP}^*[\mathbf{FHilb}_{\text{sk}}]$ which can be seen by using [26, Proposition 7.24] and [26, Proposition 7.22]. It follows that $\mathbf{CP}^*[\mathbf{FHilb}_{\text{sk}}]$ is a strict compact closed category. Note that since $\mathbf{FHilb}_{\text{sk}} \simeq \mathbf{FHilb}$ are categorically equivalent (the former is the skeleton of the latter), the results from [44] may be readily reused for the category $\mathbf{FHilb}_{\text{sk}}$ as well. It follows that $\text{Caus}[\mathbf{CP}^*[\mathbf{FHilb}_{\text{sk}}]]$ is a BV-category with negation (in the original sense) which gives us Definition 17 (1.) – (4.). Note that the authors in [44] have proven that the κ isomorphism in this category is the identity, i.e. $A^\perp \triangleleft B^\perp = (A \triangleleft B)^\perp$. Examining the constructions in [44] we see that the forgetful functor $U: \text{Caus}[\mathbf{CP}^*[\mathbf{FHilb}_{\text{sk}}]] \rightarrow \mathbf{CP}^*[\mathbf{FHilb}_{\text{sk}}]$ preserves all the required structure strictly and satisfies Definition 17 (5.) thus giving us a strong BV-category. ◀